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SARDAR PATEL UNIVERSITY
M.Sc. Information Technology (INTEGRATED) Examination
7th Semester (Reg & NC) (CBCS)
Saturday, Date: 27/10/2018
Session : Evening Time : 02:00 P.M. to 05:00 P.M.

Subject Code: PS07CIIT04

Subject Title : Advanced Computer Networking

Total Marks: 70

Q.1 Multiple Choice Questions :**[08]**

1. Sink tree is related to _____.
 - a. Static Routing
 - b. Optimality Principle
 - c. Retransmission policy
 - d. Packet Queuing
2. _____ is the responsibility of the Router.
 - a. Fill and update routing table
 - b. Detect hardware failure
 - c. Send Acknowledgement
 - d. Break packets into datagram.
3. Which of the following can be considered as a need for fragmentation?
 - a. Tunneling
 - b. Preventing packets from occupying channel too long.
 - c. Allowing packets to occupy channel too long.
 - d. All packet to pass through one network only.
4. Interior and Exterior routing differs by _____.
 - a. Path
 - b. Error rate
 - c. Storage
 - d. Cost
5. DES Consist of _____ number of stages.
 - a. 10
 - b. 16
 - c. 19
 - d. 32
6. Major concern with the security algorithms is _____ when key is to be kept secret.
 - a. Symbols used in key
 - b. Length of algorithm
 - c. Length of Key
 - d. Location of the receiver
7. _____ connection allows to establish TCP connection and send request and response and send additional request and response in pipeline.
 - a. Static
 - b. Non-persistence
 - c. Dynamic
 - d. Persistence
8. Port 25 is used by _____ protocol.
 - a. SMTP
 - b. FTP
 - c. Telnet
 - d. HTTP

Q.2 Short Questions. (Attempt any Seven)**[14]**

1. What is Region?
2. Explain Selective flooding in brief.
3. State different parameters on which accounting of data usage is done for transferring packets between two networks.
4. List different types of fragmentation.
5. State different ways to generate dynamic web pages.
6. Explain different field information stored by cookies when user access a webpage.
7. Explain S-box in brief.
8. Explain different types of bases used in Quantum Cryptography.
9. Explain DOS attack in brief.

- Q3.[a]** Explain count to infinity problem in Hierarchical routing. [06]
- Q3.[b]** Explain Shortest path routing algorithm in detail. [06]
- OR**
- Q3.[b]** Write brief note on congestion control. [06]
- Q4.[a]** State difference between Concatenated virtual circuits and datagram approach. [06]
- Q4.[b]** State the need for fragmentation. Explain transparent fragmentation with diagram. [06]
- OR**
- Q4.[b]** Explain how different networks differ. [06]
- Q5.[a]** Write short note on DSN. [06]
- Q5.[b]** Explain Email Architecture in brief. [06]
- OR**
- Q5.[b]** Draw the web model and list steps that occur on client side when a URL is requested. [06]
- Q6.[a]** Explain Substitution ciphers with example. [06]
- Q6.[b]** Explain Transposition ciphers with example. [06]
- OR**
- Q6.[b]** Explain DES in brief [06]

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